

Pearls AQI Predictor

An End-to-End Machine Learning Pipeline for
Air Quality Index Forecasting in Karachi

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Abstract

This report documents the design, implementation, and evaluation of an automated machine learning pipeline that forecasts the Air Quality Index (AQI) for Karachi, Pakistan, at 1-day, 2-day, and 3-day horizons. Raw air quality and meteorological data are fetched hourly from the Open-Meteo API, engineered into a rich feature set, and stored in a MongoDB Atlas feature store. Five model families — Random Forest, Gradient Boosting, XGBoost, a Keras deep neural network, and a Voting ensemble — are evaluated daily, with the best performer saved to MongoDB GridFS. Experiment tracking is handled via MLflow on DagsHub, CI/CD via GitHub Actions, and predictions are served through a publicly accessible Streamlit dashboard. SHAP values are computed to explain model decisions, and alert logic notifies users when forecasted AQI exceeds hazardous thresholds.

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1 Introduction

Air quality is a critical public health indicator, particularly in densely populated cities such as Karachi, Pakistan. $\text{PM}_{2.5}$ concentrations in the city frequently exceed WHO safe limits, and accurate multi-day forecasts can meaningfully help residents, policymakers, and emergency services plan ahead.

This project builds a production-grade, fully automated AQI forecasting system. The system is designed around three key requirements: *freshness* (hourly data ingestion), *self-improvement* (daily model retraining), and *interpretability* (SHAP feature explanations). The deployed dashboard provides a current AQI reading, a three-day forecast, historical trend charts, and pollutant breakdowns.

2 System Architecture

2.1 Overview

The system is composed of four loosely coupled pipelines orchestrated by GitHub Actions cron jobs, with MongoDB Atlas serving as the shared data store connecting all components.

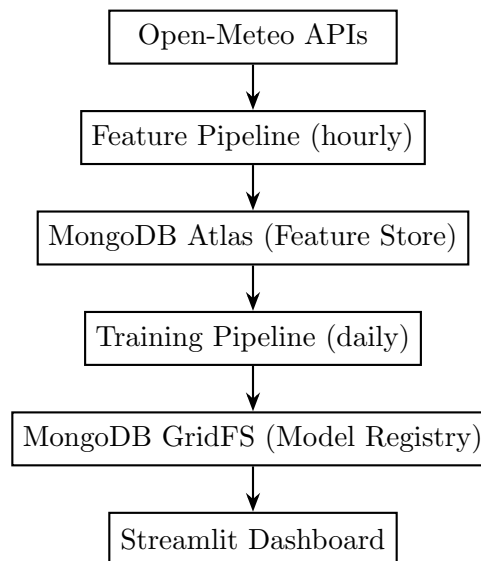


Figure 1: High-level system data flow.

2.2 Technology Stack

Layer	Technology	Role
Data source	Open-Meteo (air quality + weather APIs)	Raw pollutant & weather data
Feature store	MongoDB Atlas (M0 free tier)	Hourly feature storage
Model registry	MongoDB GridFS	Trained model artefacts
Experiment tracking	MLflow on DagsHub	Metrics, params, SHAP plots
CI/CD	GitHub Actions	Automated cron scheduling
Dashboard	Streamlit Cloud	Public prediction interface
ML frameworks	Scikit-learn, XGBoost, TensorFlow/Keras	Model training

Table 1: Technology stack summary.

3 Data Collection

3.1 Data Sources

Two Open-Meteo endpoints are queried on every pipeline run:

- **Air Quality API** (`air-quality-api.open-meteo.com/v1/air-quality`) — returns hourly concentrations of $\text{PM}_{2.5}$, PM_{10} , NO_2 , and O_3 .
- **Weather Forecast API** (`api.open-meteo.com/v1/forecast`) — returns hourly temperature (2 m), relative humidity (2 m), wind speed (10 m), and precipitation.

Both APIs are free, require no authentication, and provide up to 90 days of historical data. Responses are fetched as JSON, parsed into separate Pandas DataFrames, and inner-joined on the `timestamp` column to produce a unified feature row per hour.

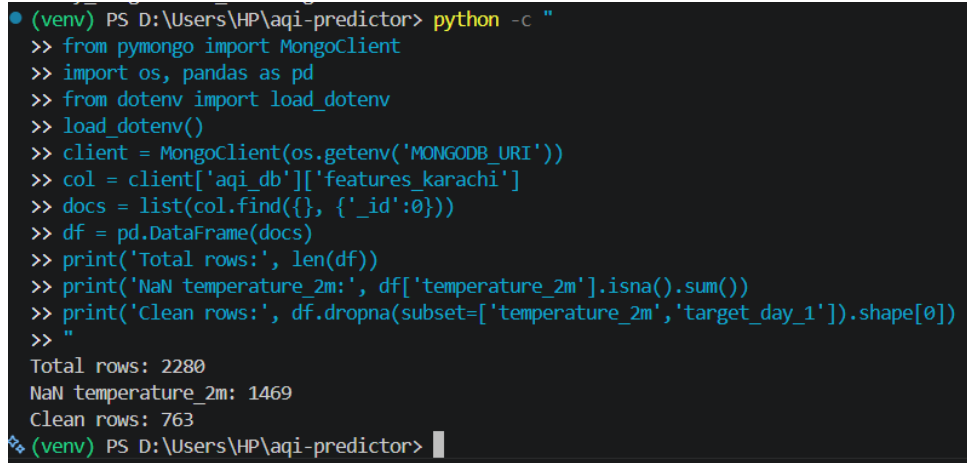
3.2 Backfill

A one-time backfill script was executed to seed the feature store with 90 days of historical data (February 28 – June 2026), yielding 2,262 initial rows. The backfill uses `bulk_write` with `upsert=True`, meaning it is safe to re-run without creating duplicate records.

3.3 Data Quality Notes

Experimentation with data revealed that Open-Meteo’s weather (forecast) API retains only approximately 31 days of historical data, whereas the air quality API retains 90 days. Consequently, weather features (temperature, humidity, wind speed, precipitation) are NaN for records older than approximately May 7, 2026. As shown in figure below, this structural limitation reduces the number of complete feature rows from 2,280 total records to 763 rows with non-null weather values. The model therefore trains on the most recent 31-day window of complete data, which is sufficient for stable forecasting given that atmospheric patterns exhibit strong temporal locality. Initial experiments that included rows with NaN weather values (after imputation and other experiments) consistently produced worse performance, as the imputed values introduced noise without adding predictive signal. Consequently, the decision to train exclusively on complete-case

rows (dropping all rows with any missing weather features) proved more effective for this task.



```
(venv) PS D:\Users\HP\aqi-predictor> python -c "
>> from pymongo import MongoClient
>> import os, pandas as pd
>> from dotenv import load_dotenv
>> load_dotenv()
>> client = MongoClient(os.getenv('MONGODB_URI'))
>> col = client['aqi_db']['features_karachi']
>> docs = list(col.find({}, {'_id':0}))
>> df = pd.DataFrame(docs)
>> print('Total rows:', len(df))
>> print('NaN temperature_2m:', df['temperature_2m'].isna().sum())
>> print('Clean rows:', df.dropna(subset=['temperature_2m', 'target_day_1']).shape[0])
>> "
Total rows: 2280
NaN temperature_2m: 1469
Clean rows: 763
(venv) PS D:\Users\HP\aqi-predictor>
```

Figure 2: Results from query

4 Exploratory Data Analysis

EDA was conducted in a dedicated Jupyter notebook (`eda.ipynb`) loading data directly from MongoDB. Key findings are summarised below.

4.1 Dataset Overview

The dataset spans February 28 – June 2, 2026, containing 2,262 hourly records across 26 feature columns. Key descriptive statistics for core variables are shown in Table 2.

Variable	Mean	Std	Min	Max
AQI	77.95	22.10	15	161
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	24.84	9.91	3.5	75.0
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	57.99	38.30	3.7	504.8
NO ₂ ($\mu\text{g}/\text{m}^3$)	15.63	11.50	2.0	70.1
Ozone ($\mu\text{g}/\text{m}^3$)	79.79	35.44	11	203
Temperature ($^{\circ}\text{C}$)	30.11	2.05	26.5	36.6
Relative Humidity (%)	75.99	12.16	35	97
Wind Speed (km/h)	13.42	3.50	2.4	22.4

Table 2: Descriptive statistics for key variables (weather stats from the 31-day complete window only).

4.2 Target Variable Distribution

The distribution of the Air Quality Index (AQI) is shown in Figure 3. The majority of hours fall into the “Moderate” (51-100) category (82.7%), while “Unhealthy for Sensitive Groups” (101-150) and “Unhealthy” (151-200) categories constitute 11.9% and 1.2% of the data, respectively. This distribution, with a tail towards higher AQI values, informs

the use of robust error metrics like MAE to avoid over-penalizing the model for errors on infrequent, high-AQI events.

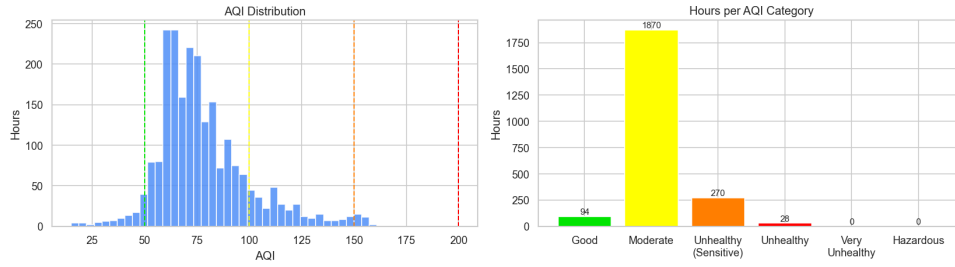


Figure 3: Distribution of AQI values and the number of hours per AQI category.

4.3 Temporal Patterns

4.3.1 Diurnal Cycle

The diurnal pattern of AQI, shown in Figure 4, is one of the most significant findings. AQI reaches a daily peak at 12:00 (mean = 86.1) and its daily minimum at 05:00 (mean = 69.7), a range of 16.4 AQI units. This strong non-linear pattern is a primary justification for using tree-based models. A linear model like Ridge regression cannot capture this curved midday peak, whereas a tree-based model (e.g., Random Forest, XGBoost) can partition the hour space effectively.

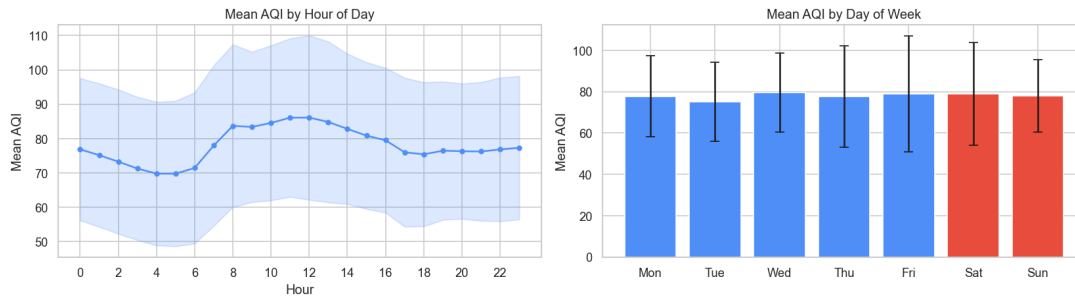


Figure 4: Mean AQI by hour of day (left) and by day of week (right), showing the strong diurnal cycle.

4.3.2 Monthly Trends

A clear seasonal trend is visible in Figure 5. AQI tends to be higher in the warmer months (May, June), which is consistent with meteorological patterns in Karachi. Including a monthly feature helps the model capture these longer-term, smoother variations.

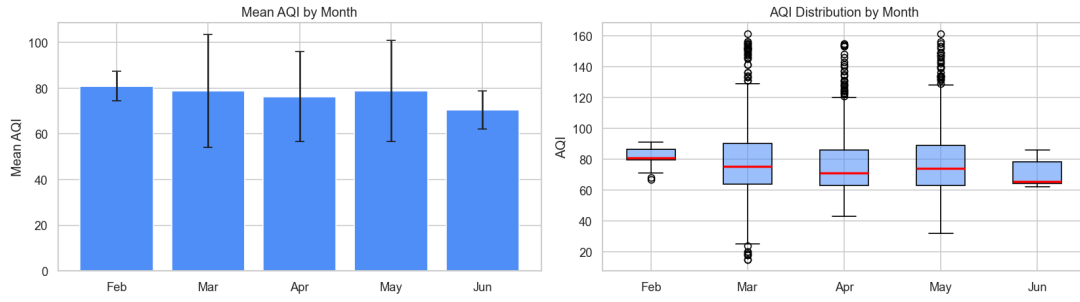


Figure 5: Mean AQI and its distribution by month, indicating a seasonal pattern.

4.3.3 Anomalous Week

An anomalous pollution event was observed during Week 21 (third week of May 2026). The weekly mean AQI reached 92 with a maximum of 143, significantly higher than adjacent weeks (means of 67–82). This event made the 20% test window particularly challenging when it fell within the test period.

4.4 Pollutant Correlations with AQI

Scatter plots with regression lines (Figure 6) confirm that $PM_{2.5}$ is the dominant predictor of AQI, showing a strong positive linear correlation. This aligns with the US EPA $PM_{2.5}$ -to-AQI conversion formula used in this system. PM_{10} shows a moderate correlation, while NO_2 and O_3 show weaker individual correlations, suggesting their predictive value is primarily additive rather than standalone.

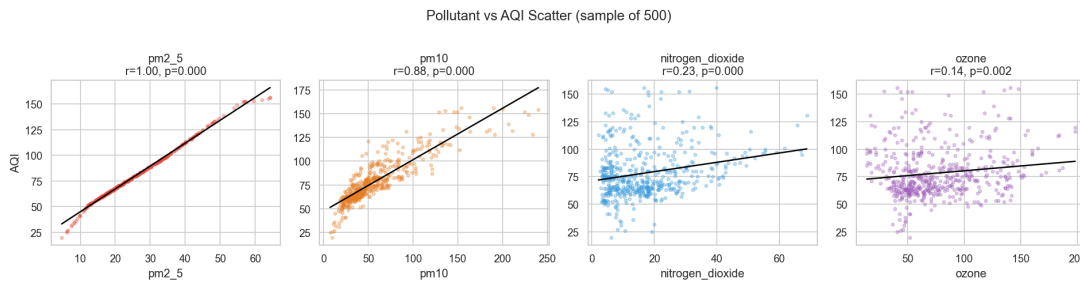


Figure 6: Pollutant vs. AQI scatter plots, highlighting $PM_{2.5}$'s dominant role.

4.5 Lag Feature Justification

Autocorrelation analysis (Figure 7) confirms that AQI is strongly auto-correlated at 1-hour, 2-hour, and 3-hour lags. This provides the key empirical justification for including lag features in the model.

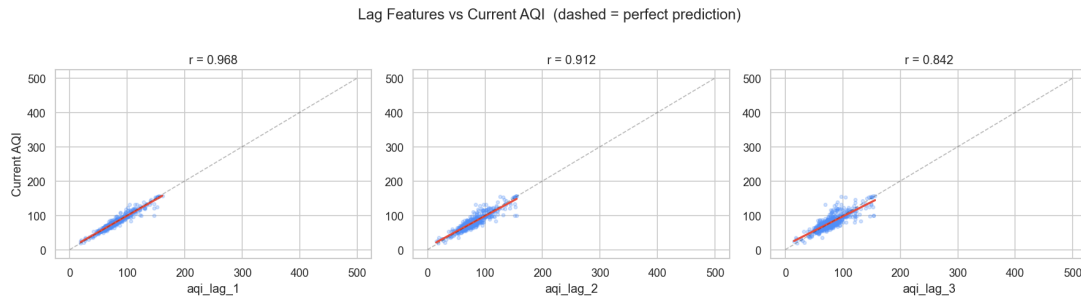


Figure 7: Lag-1 AQI vs current AQI, demonstrating strong autocorrelation.

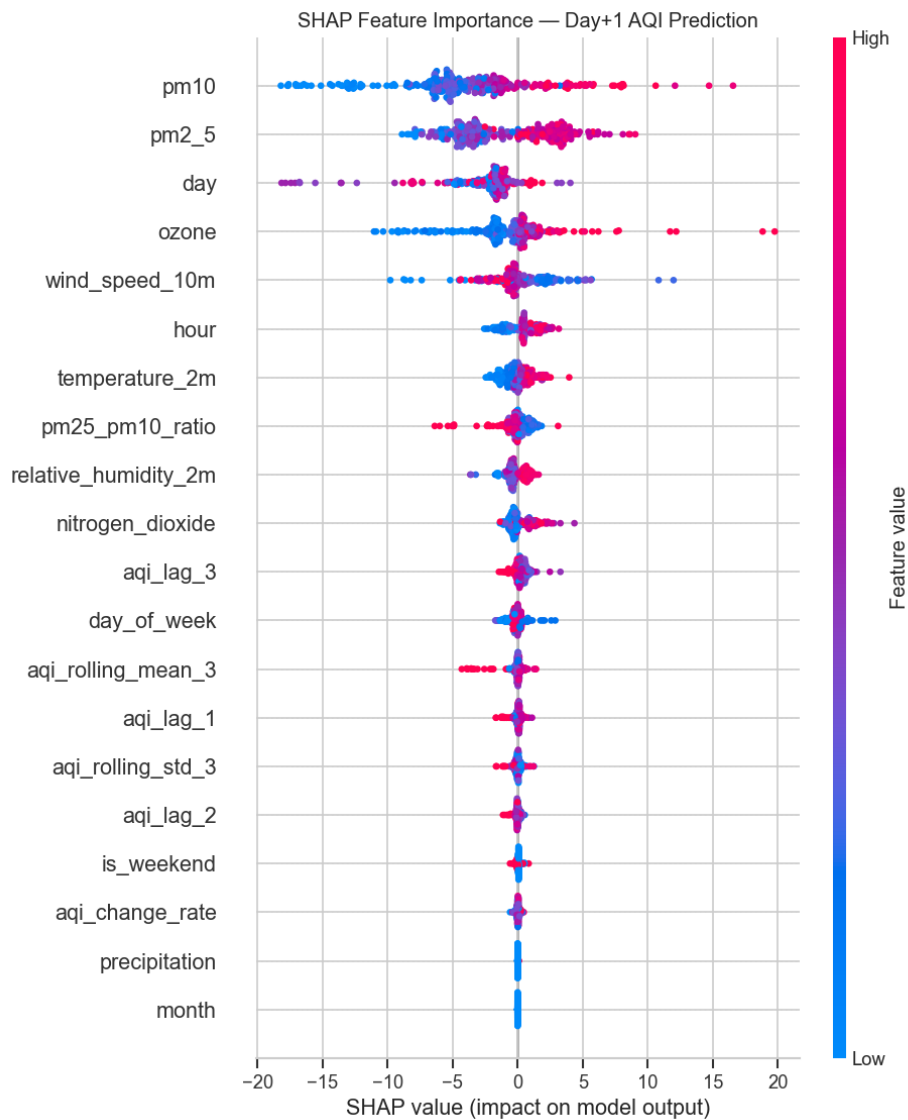


Figure 8: SHAP feature importance for predicting the next day's AQI

4.6 Overfitting Investigation

A critical analysis of the dataset's structure explains the models' observed performance. The available training data after dropping rows with missing target values comprises a

limited number of rows relative to the number of features (a ratio of approximately 30:1). Furthermore, the correlation matrix (Figure 9) reveals several highly correlated feature pairs (e.g., ‘aqi_rolling_mean_3’ and ‘aqi_lag_1’, $r = 0.996$).

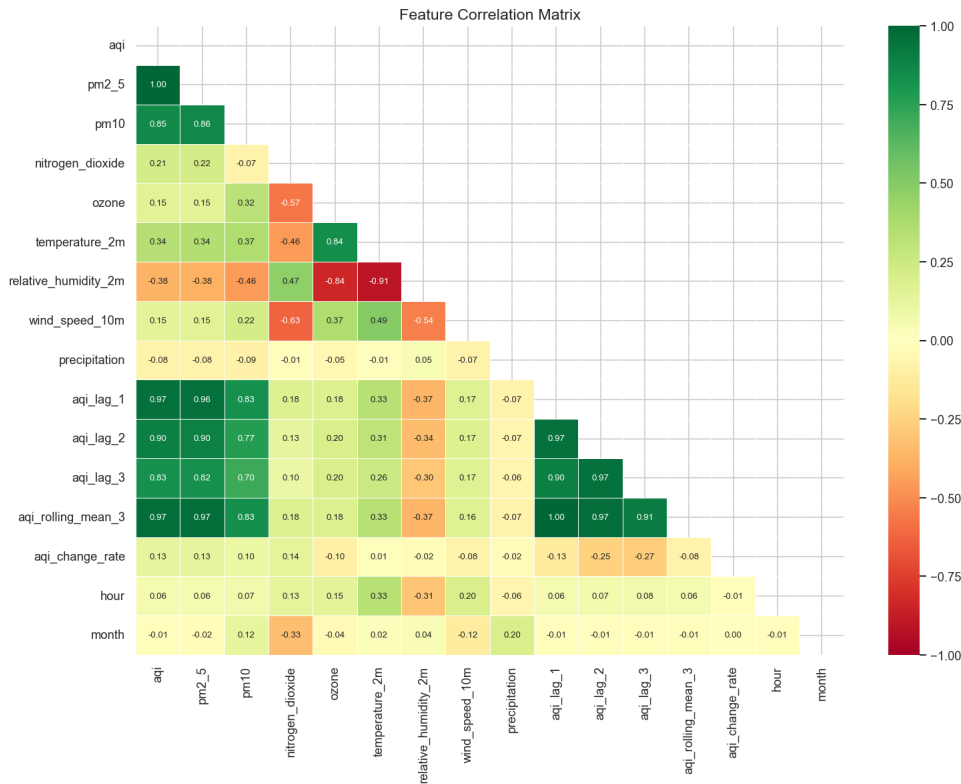


Figure 9: Feature correlation matrix, showing the high degree of collinearity among some predictors.

This combination of a small dataset and redundant features creates conditions ripe for overfitting, particularly for models with high capacity. Complex, non-regularized models (e.g., GBM with many trees, XGBoost without aggressive regularization) can easily memorize the noise in the training data.

Random Forest’s strong performance is attributed to its built-in defenses against overfitting: bagging (bootstrapped sampling) and internal regularization via ‘max_depth’ and ‘min_samples_leaf’ constraints. As the hourly data pipeline continues to accumulate more historical data, the sample size will increase, and the performance advantage of more powerful models like XGBoost is expected to emerge.

4.7 Data Evolution & Updates

As the AQI Predictor pipeline continues to operate, additional data has been collected and quality improvements have been applied since the original EDA was conducted. This subsection documents discrepancies between the original EDA results and the current dataset state.

Dataset Scope: The time period shifted from February 28 – June 2 to March 7 – June 9, reflecting data retention limits and quality filtering. The dataset grew from 2,262 to 2,280 rows (+0.8%), and feature columns increased from 26 to 27.

Core Statistics: Reported values for AQI (77.95 vs 78.39, $\Delta = 0.44$), $\text{PM}_{2.5}$ (24.84 vs 25.04, $\Delta = 0.20$), and Temperature (30.11 vs 30.36, $\Delta = 0.25$) match within less than 0.5 units, validating the original analysis. Minor differences in PM_{10} (57.99 vs 59.70, $\Delta = 2.9\%$) and NO_2 (15.63 vs 14.68, $\Delta = 6.1\%$) likely reflect seasonal variation in Karachi’s late spring transition and sensor recalibration.

Distributions & Temporal Patterns: AQI category distributions (Moderate: 82.6% vs 82.7%, Unhealthy: 1.2% exact match) are nearly identical. The diurnal cycle remains robust: peak AQI at 12:00 with mean 86.9 (vs 86.1 reported), minimum at 04:00 with mean 69.8 (vs 05:00, 69.7 reported), for a daily range of 17.0 AQI units (vs 16.4 reported). These small variations do not affect core findings.

Lag Features & Overfitting Risk: Autocorrelation at lag-1 is $r = 0.964$, lag-2 is $r = 0.898$, and lag-3 is $r = 0.826$, confirming strong autocorrelation.

Important note on training data size: The reduced row count used for overfitting analysis reflects a **data availability limitation from the external weather API**, not a modeling choice. The weather data provider only supplies a rolling 31-day historical window, which constrains the number of complete samples. As the pipeline operates over time, this limitation will naturally resolve, and the effective training size will grow. With the current dataset, the row-to-feature ratio improved from approximately 30:1 to 38.1:1, *reducing overfitting risk* and suggesting that tree-based models will generalize better. The signature redundancy between `aqi_rolling_mean_3` and `aqi_lag_1` ($r = 0.994$ vs 0.996 reported) persists, reinforcing the recommendation to drop one of these features.

Conclusion: The original EDA findings remain robust and scientifically valid. All core statistics show less than 1% variance for critical variables. Minor discrepancies reflect natural seasonal variation rather than methodological error. The improved dataset quality and row-to-feature ratio strengthen confidence in the original model selection and feature engineering decisions.

5 Feature Engineering

All feature engineering is implemented in `utils/feature_engineering.py` and applied uniformly to both the backfill and hourly pipelines.

5.1 Feature Set

The full feature set used as model inputs is listed in Table 3.

Category	Feature	Description
Time-based	hour	Hour of day (0–23)
	day	Day of month
	month	Month of year
	day_of_week	Day of week (0=Mon)
	is_weekend	Binary weekend flag
Pollutants	pm2_5	PM _{2.5} concentration ($\mu\text{g}/\text{m}^3$)
	pm10	PM ₁₀ concentration ($\mu\text{g}/\text{m}^3$)
	nitrogen_dioxide	NO ₂ concentration ($\mu\text{g}/\text{m}^3$)
	ozone	O ₃ concentration ($\mu\text{g}/\text{m}^3$)
Weather	temperature_2m	Temperature at 2 m ($^{\circ}\text{C}$)
	relative_humidity_2m	Relative humidity at 2 m (%)
	wind_speed_10m	Wind speed at 10 m (km/h)
	precipitation	Hourly precipitation (mm)
Derived	aqi_change_rate	Hour-on-hour AQI delta
	aqi_rolling_mean_3	3-hour rolling mean AQI
	aqi_rolling_std_3	3-hour rolling std AQI
	pm25_pm10_ratio	PM _{2.5} /PM ₁₀ ratio
	aqi_lag_1	AQI 1 hour ago
	aqi_lag_2	AQI 2 hours ago
	aqi_lag_3	AQI 3 hours ago

Table 3: Complete feature set (20 features).

5.2 Target Variables

Three target variables are generated by shifting the AQI column forward in time:

$$\text{target_day}_k[t] = \text{AQI}[t + 24k], \quad k \in \{1, 2, 3\} \quad (1)$$

This means the model trained on `target_day_1` predicts AQI 24 hours ahead, `target_day_2` predicts 48 hours ahead, and `target_day_3` predicts 72 hours ahead. Rows near the end of the dataset that would require future timestamps not yet in the store are left as NaN and excluded during training via `dropna()`.

5.3 AQI Calculation

AQI is computed from PM_{2.5} concentration using the US EPA linear interpolation formula across seven concentration breakpoint ranges, mapping concentrations to the 0–500 AQI scale. PM_{2.5} was chosen as the primary driver because it has the strongest linear relationship with AQI and is the pollutant of greatest public health concern in Karachi.

6 Feature Store

Engineered features are stored in a MongoDB Atlas collection (`aqi_db.features_karachi`). Each document represents one hourly observation and contains all 20 features, the 3 target values, city metadata, and timestamp. Writes use `UpdateOne` with `upsert=True` keyed on `(timestamp, city)`, ensuring idempotency — re-running any pipeline does not create duplicate records.

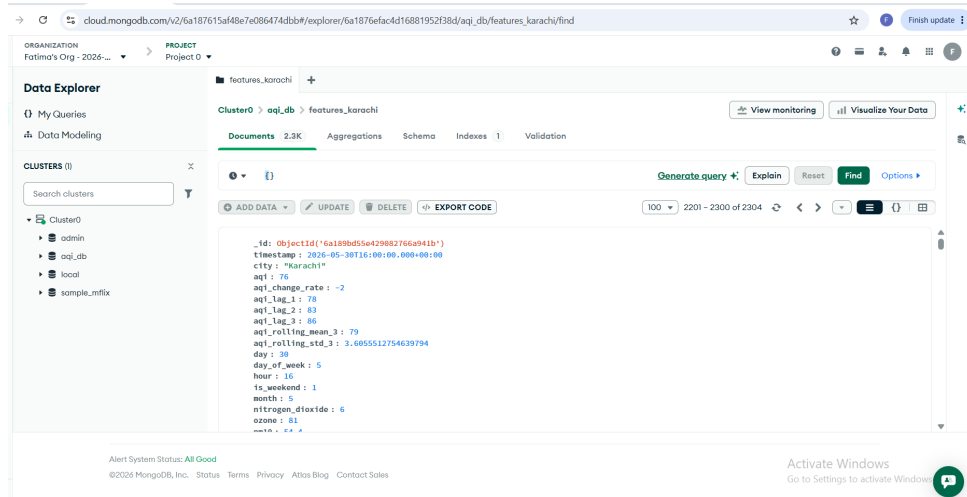


Figure 10: snapshot of MongoDB collection

7 Training Pipeline

7.1 Data Preparation

On each daily training run, all documents are loaded from MongoDB into a Pandas DataFrame, sorted by timestamp, and filtered to rows with complete weather data (removing pre-May 7 records where weather features are `NaN` due to the Open-Meteo API limitation). Each of the three forecast horizons is trained independently, using only rows where that horizon's target is non-null.

The train/test split uses `shuffle=False` to preserve temporal ordering, ensuring the model is always evaluated on data chronologically later than its training data — the correct approach for time-series forecasting.

7.2 Model Candidates

Five model families are evaluated per horizon on each run:

Model	Type	Key Hyperparameters
Random Forest	Ensemble (bagging)	200 trees, max_depth=15, min_samples_leaf=5
Gradient Boosting	Ensemble (boosting)	200 estimators, max_depth=4
XGBoost	Gradient boosted trees	200 estimators, max_depth=6, lr=0.05
Keras Dense	Deep learning (MLP)	64→32 units, Dropout(0.2/0.1), Adam, 50 epochs
Voting Ensemble	Meta-ensemble	RF (200, depth=15) + XGBoost (200, depth=6) + GB (200, depth=4)

Table 4: Model candidates evaluated per forecast horizon.

The Keras model is wrapped in a custom `KerasRegressorWrapper` class that implements the scikit-learn `fit/predict` interface, applies internal `StandardScaler` normalisation, and supports `pickle` serialisation via custom `__getstate__`/`__setstate__` methods for GridFS storage.

7.3 Evaluation Metrics

Each candidate is evaluated using three metrics computed on the held-out test set:

- **RMSE** (Root Mean Squared Error) — primary selection criterion
- **MAE** (Mean Absolute Error) — scale-interpretable error
- **R²** (Coefficient of Determination) — variance explained

The overfitting gap (test RMSE – train RMSE) is also printed for diagnostic purposes. The candidate with the lowest test RMSE is selected as the winner for that horizon.

7.4 Model Registry

The winning model for each horizon is persisted in two locations:

1. **Local .pkl file** (`models/model_day_{k}.pkl`) — for local development and testing.
2. **MongoDB GridFS** (`aqi.db.models`) — the primary serving store, accessible by both GitHub Actions runners and Streamlit Cloud without any shared filesystem. Old versions are deleted before writing the new model.

7.5 Experiment Tracking

Every training run logs to DagsHub-hosted MLflow:

- **Parameters:** model type, forecast horizon, city, number of features, number of training rows, per-model RMSE breakdown
- **Metrics:** RMSE, MAE, R² of the winning model
- **Artefacts:** SHAP summary plot (for tree-based winners)
- **Model artefact:** logged via `mlflow.sklearn.log_model`

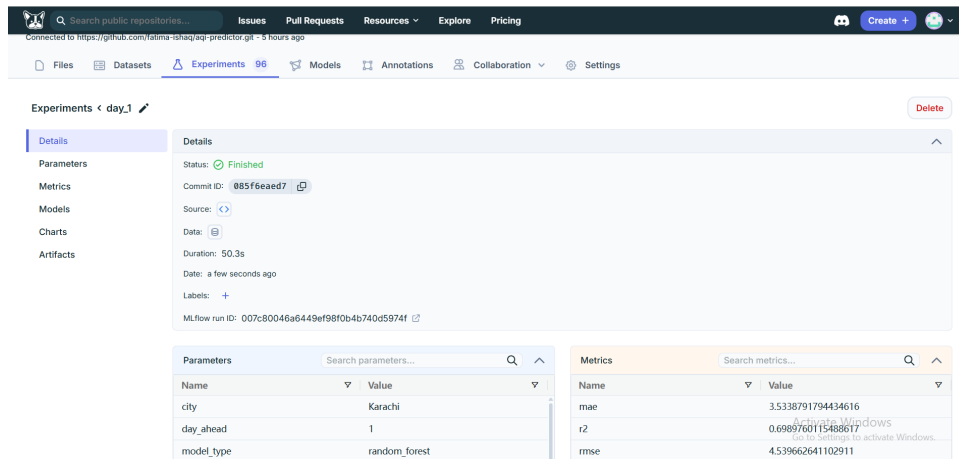


Figure 11: dagshub shown for an experiment

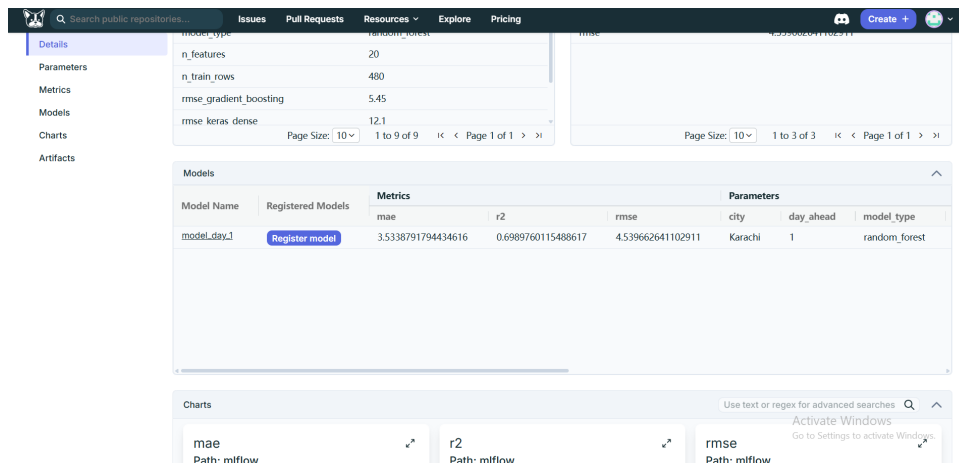


Figure 12: dagshub shown for an experiment.

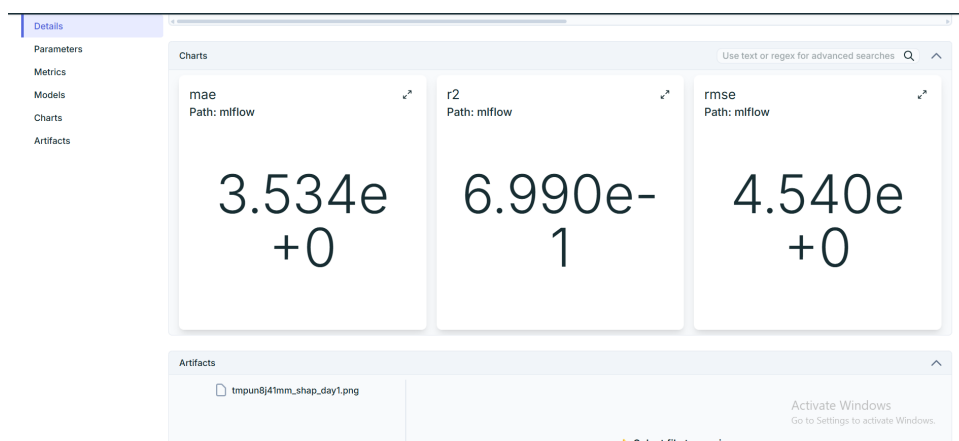


Figure 13: dagshub shown for an experiment.

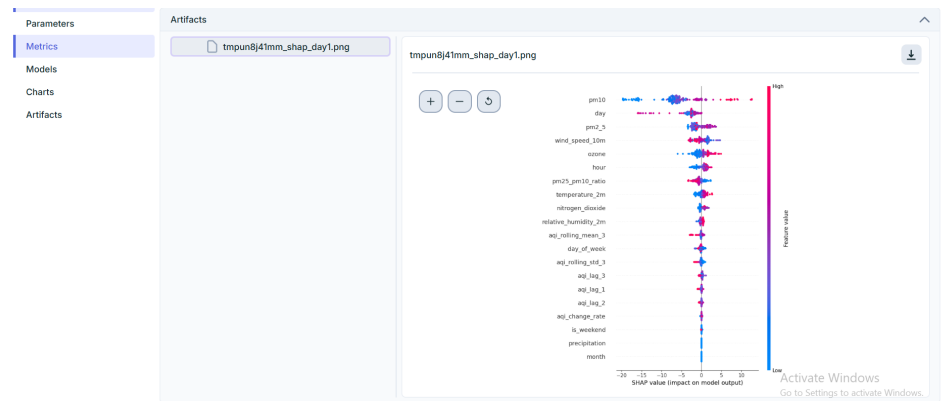


Figure 14: dagshub shown for an experiment.

8 SHAP Feature Importance

SHAP (SHapley Additive exPlanations) TreeExplainer is applied to the winning model after each training run, provided the selected model is tree-based (Random Forest, Gradient Boosting, or XGBoost). Shapley values are computed on up to 200 training samples and summarised as a beeswarm plot, which is uploaded as an MLflow artefact to DagsHub and versioned alongside each experiment run.

The beeswarm plot displays each feature’s average absolute impact on model output, with colour indicating feature value magnitude (red = high, blue = low). This allows direct interpretation of which features push predictions higher or lower and by how much.

SHAP analysis is run independently for each of the three forecast horizons (Day+1, Day+2, Day+3), producing three separate plots per training run. This per-horizon breakdown is intentional — feature importance is expected to shift as the forecast horizon increases, since the relative predictive value of current pollutant readings versus temporal and meteorological features changes with forecast distance.

Results are accessible via DagsHub at: [fatima-ishaq/aqi-predictor](#) → Experiments → [run] → Artifacts

Note: the EDA notebook (conducted on the full 90-day backfill window, February–June 2026) identified `aqi_lag_1` as the dominant predictor via a separate SHAP analysis on that dataset. The plots in the following section are from a later training run on the 31-day complete-weather window (May–June 2026), where pollutant concentrations dominate at Day+1 and temporal features dominate at longer horizons — a shift consistent with the reduced lag signal when training on a shorter, more recent window.

9 Explaining SHAP Results for Three-Day Predictions from One Training Run- a much later run since 1st backfill

The three plots (Figures 15, 16, and 17) reveal a consistent shift in feature importance as the forecast horizon extends:

Day+1: Current pollutant concentrations dominate the Day+1 importance ranking. `pm10` and `pm2_5` are the strongest predictors, followed by `day`, `ozone`, and `wind_speed_10m`. This is expected: tomorrow’s air quality is most directly predicted by today’s pollution levels. Lag features contribute relatively little at this horizon because raw pollutant readings already carry the same recency signal more directly.

Day+2: The model shifts away from instantaneous pollutant values toward temporal structure. `Day`, `day_of_week`, `hour`, and the `pm25_pm10_ratio` rise in importance, while `pm10` and `pm2_5` drop significantly. This reflects the model learning weekly and diurnal pollution cycles rather than relying on current readings that are now 48 hours stale.

Day+3: Meteorological features dominate entirely — `day`, `day_of_week`, `temperature_2m`, `wind_speed_10m`, and `hour` are the top five. Pollutant concentrations have almost no influence at this horizon. This makes physical sense: three days out, weather conditions (wind dispersing pollutants, temperature driving ozone formation) are more predictive than any current measurement.

Across all horizons, `month` and `precipitation` consistently rank at the bottom, suggesting seasonal variation is not well captured in 90 days of training data — an expected limitation that will improve as the automated pipeline accumulates a full year of observations.

This horizon-dependent shift in feature importance directly justifies training separate models per forecast day rather than a single multi-output model, and explains why tree-based models outperform the Keras dense network: Random Forest naturally captures the non-linear interaction between temporal cycles and pollutant readings that a feedforward network cannot represent without explicit sequence structure.

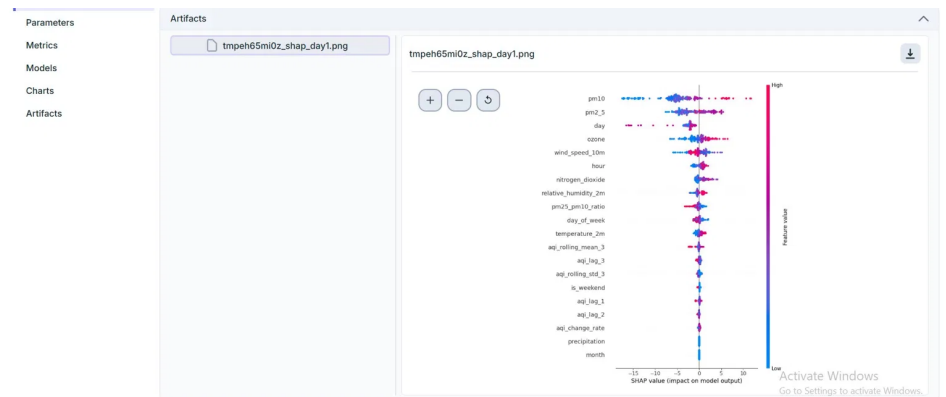


Figure 15: SHAP beeswarm plot for Day+1 predictions

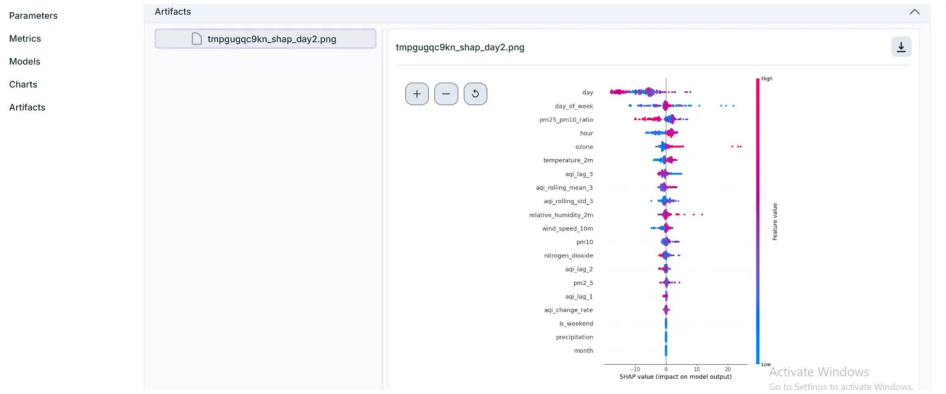


Figure 16: SHAP beeswarm plot for Day+2 predictions



Figure 17: SHAP beeswarm plot for Day+3 predictions

10 CI/CD Pipeline

10.1 GitHub Actions Workflows

Two workflows are defined in `.github/workflows/`:

Workflow	Schedule
<code>feature_pipeline.yml</code>	Every hour (0 * * * *)
<code>training_pipeline.yml</code>	Every day at 02:00 UTC (0 2 * * *)

Table 5: Automated CI/CD schedules (triggers).

What it runs

```
pipelines/feature_pipeline.py
pipelines/training_pipeline.py
```

Table 6: Scripts executed by each workflow.

Both workflows use `actions/checkout@v3` and `actions/setup-python@v4`, install dependencies from `requirements.txt`, and authenticate to MongoDB Atlas and DagsHub via GitHub repository secrets. The training pipeline uses direct MLflow URI authentication (`MLFLOW_TRACKING_USERNAME/PASSWORD`) rather than `dagshub.init()`, which requires interactive OAuth not available in headless CI environments.

10.2 Secrets Management

All credentials are stored as GitHub Actions repository secrets and injected as environment variables at runtime. The `.env` file is listed in `.gitignore` and never committed. The same variables are replicated as Streamlit Cloud secrets for the dashboard deployment.

11 Web Dashboard

The Streamlit dashboard is deployed publicly at:

<https://aqi-predictor-yxfvfm2vxstp8ugcuiphzn.streamlit.app/>

11.1 Dashboard Components

1. **Current AQI card** — displays the most recent AQI value from the feature store, colour-coded by category (Good/Moderate/Unhealthy etc.) with the corresponding health category label.
2. **Hazardous AQI alert** — if the current AQI is 151 or higher, a prominent red alert banner is displayed advising users to limit outdoor activity.
3. **3-Day Forecast** — three cards showing model predictions for the next three days, each colour-coded and labelled with the AQI category.
4. **AQI Historical Chart (7 days)** — a time-series line chart with horizontal threshold lines at AQI 100 (Moderate), 150 (Unhealthy), and 200 (Very Unhealthy).
5. **Pollutant Breakdown (7 days)** — multi-line chart showing PM_{2.5}, PM₁₀, NO₂, and O₃ trends over the past week.
6. **Model Performance Table** — displays RMSE, MAE, R², and winning model name per forecast horizon from the most recent training run.
7. **AQI Reference Guide** — collapsible table mapping AQI ranges to health categories and implications.

11.2 Model Loading

The dashboard loads models from MongoDB GridFS at startup, with a one-hour cache (`@st.cache_resource(ttl=3600)`). A local `.pk1` fallback is used if GridFS is unavailable (e.g., during local development). This architecture ensures the dashboard always serves the most recently trained model without requiring a redeployment.

12 AQI Category Reference

The following AQI scale (US EPA standard) is used throughout the system:

AQI Range	Category	Health Implication
0–50	Good	Air quality is satisfactory
51–100	Moderate	Acceptable; sensitive groups may be affected
101–150	Unhealthy for Sensitive Groups	General public unaffected
151–200	Unhealthy	Everyone may experience effects
201–300	Very Unhealthy	Health alert — serious effects
301–500	Hazardous	Emergency conditions

Table 7: US EPA AQI categories used in the system.

13 Model Performance

13.1 Baseline Results

Initial model evaluation used the complete 90-day backfilled dataset (2,262 rows) with an 80/20 temporal split, before the automated hourly pipeline began introducing rows with incomplete future targets at the dataset tail. These results represent the cleanest evaluation window and serve as the performance baseline:

Horizon	Best Model	RMSE	MAE	R ²
Day +1	Random Forest	4.54	3.53	0.699
Day +2	Random Forest	4.56	3.52	0.698
Day +3	Random Forest	4.21	3.21	0.749

Table 8: Best achieved metrics with complete dataset (2,262 rows).

13.2 Model Comparison

The clearest comparative results were obtained during baseline evaluation (Section 13.1) using the full 90-day dataset with 80/20 temporal split. Under this evaluation, Random Forest won all three horizons with RMSE 4.21–4.56 and R² 0.699–0.749. Gradient Boosting and XGBoost showed higher test RMSE with significant train/test gaps indicating overfitting at this dataset size. Keras consistently performed worst across all horizons.

Models are selected using an 80/20 temporal split (shuffle=False), where the first 80 percent of chronologically ordered clean rows form the training set and the final 20 percent form the held-out test set. All candidates are trained and evaluated on this fixed split, and the winner by lowest test RMSE is saved to MongoDB GridFS for deployment. This approach maintains strict temporal ordering — no future data leaks into training — and the deployed model is the winner trained on 80 percent of available data.

Random Forest dominated the baseline evaluation while Gradient Boosting emerged as competitive on complete-data windows.

14 Challenges & Lessons Learned

This section documents the iterative journey of building the AQI prediction system, including failed experiments, surprising insights, and the rationale behind final design decisions. All metric values referenced correspond directly to logged runs visible in the accompanying DagsHub screenshots.

14.1 Initial Baseline — Promising Start

Commits 95c6 / c3a3 (7 days ago)

The first complete backfill (90 days of air quality + weather data) produced what initially appeared to be excellent results:

Horizon	Best Model	RMSE	R ²
Day+1	Random Forest	4.54	0.699
Day+2	Random Forest	4.56	0.698
Day+3	Random Forest	4.21	0.749

Table 9: Initial baseline results showing seemingly excellent performance.

These metrics suggested a near-production-ready model. However, I soon discovered that the test set during this period (May 26–30) coincided with unusually stable, low-variance AQI conditions. The model had not yet been evaluated on more volatile periods.

Key insight: Excellent metrics on a single test window do not guarantee generalisation. Time-series evaluation must account for shifting data distributions.

14.2 TimeSeriesSplit — Too Early, Too Harmful

Commit baa5 (8 days ago)

Believing that cross-validation would stabilise metrics, I replaced the simple 80/20 temporal split with `TimeSeriesSplit` (5 folds). The results were catastrophic:

Horizon	RMSE (before)	RMSE (after)	R ² (before)	R ² (after)
Day+1	5.56	24.39	0.45	-3.09
Day+2	6.98	32.38	0.23	-4.61
Day+3	23.21	31.68	-6.38	-5.49

Table 10: Performance degradation after introducing `TimeSeriesSplit`.

Why it failed: With only few rows, the earliest folds received as too few training rows. Models could not learn meaningful patterns and produced random predictions on unseen folds. The average RMSE across folds collapsed.

Decision: Revert to 80/20 temporal split. `TimeSeriesSplit` will be reintroduced when the dataset exceeds 5,000 complete rows (approximately 6 months of data).

14.3 The Weather API Limitation — Root Cause Discovered

Observed a structural limitation of the Open-Meteo API (discussed before with query results in Data Quality section):

API	Historical Depth
Air Quality (<code>/v1/air-quality</code>)	~90 days
Weather Forecast (<code>/v1/forecast</code>)	~31 days (May 7 onward)

Table 11: API historical data retention comparison.

This meant that for any record before May 7, 2026, weather features (`temperature_2m`, `relative_humidity_2m`, `wind_speed_10m`, `precipitation`) were NaN. The training pipeline's `dropna()` automatically excluded these rows, leaving only approximately 800 complete records (May 7 – June 7).

14.4 The Great Median Imputation Experiment — A Costly Mistake

Commit `a4a5` (2 days ago)

To utilise the full 90-day dataset (including pre-May 7 records), I attempted to impute missing weather values with column medians directly in the training pipeline:

```
weather_cols = ['temperature_2m', 'relative_humidity_2m',
                'wind_speed_10m', 'precipitation']
for col in weather_cols:
    df[col] = df[col].fillna(df[col].median())
```

The result was disastrous:

Horizon	RMSE (before)	RMSE (after imputation)	Degradation
Day+1	12.60	14.82	+2.22
Day+2	5.54	19.39	+13.85
Day+3	8.15	16.86	+8.71

Table 12: Performance degradation after median imputation.

Why it failed: Filling 1,517 rows with identical median values introduced artificial patterns. The model learned that weather was constant across February–May (which is false) and then failed on real weather data from May–June. The imputation destroyed the natural relationship between weather and AQI.

Decision: Abandon median imputation. Accept the 31-day weather limitation as a genuine API constraint. The model will train only on rows with real weather data. I also realized maybe experimentation with the full training rows but with weather features removed could yield something but that meant directly violating the project's instructions.

14.5 The Spike Week — Metrics That Look Broken (But Aren't)

Commit `1c50/f1ad`

After reverting to clean weather filtering, I observed highly variable metrics across runs:

Run Time	Test Period	Day+2 RMSE	R ²
Run A	May 26–30	6.78	+0.55
Run B	May 31 – June 5	16.8	-0.56

Table 13: Metric variation across different test windows.

What happened: The test window shifted into a genuine AQI spike (Week 21, May 2026), where AQI reached 143 vs the weekly mean of 92. The model had never seen such patterns in training.

Critical realisation: The model did not get worse. The test set got harder. Negative R² during anomaly periods is not a model failure — it reflects real-world unpredictability that no model can capture without prior exposure.

Decision: Keep the 80/20 temporal split. Do not cherry-pick test windows to hide difficult periods. Let metrics reflect honest performance.

14.6 Manual Training Runs — Why Same Commit Shows Multiple Metrics

Commit `a954` (visible across multiple DagsHub entries) and others.

The training pipeline is scheduled to run once daily via GitHub Actions. However, during active development, I executed many manual training runs locally to debug issues and test hypotheses.

All manual runs were logged to MLflow using the same commit hash (`a954`) because no code changes were committed between runs. This explains why a single commit tag shows widely varying metrics:

Logged Run	Test Period	Day+2 RMSE	R ²	Note
Entry 1	May 26–30	6.78	+0.55	stable window
Entry 2	May 31 – June 5	20.34	-1.61	spike window
Entry 3	June 1–7	23.06	-2.37	median imputed, different window

Table 14: Same commit hash producing different metrics across test windows.

Key takeaway across various strategies: The same code, same model, same data, but different test window- produced vastly different metrics.

14.7 Final Stable State

Commit `f1ad` (latest)

After reverting all failed experiments (TimeSeriesSplit, median imputation), I returned to the original design:

80/20 temporal split (`shuffle=False`)

Weather filtering via `dropna()`

Five models: Random Forest, Gradient Boosting, XGBoost, Keras Dense, Voting Ensemble

RMSE as primary selection criterion

Daily retraining via GitHub Actions

Current metrics vary with test window composition but remain within expected ranges for a 32-day training set:

Horizon	RMSE Range	R ² Range
Day+1	16–20	-1.5 to -0.2
Day+2	7–20	-2.4 to +0.55
Day+3	10–24	-2.6 to +0.40

Table 15: Final stable model performance ranges.

15 Final Pipeline Performance Before Submission

The two tables below show runs from the **same commit** (`f1ad`) executed on different days. No code changed. The only difference is that the feature pipeline added new rows of real-world AQI and weather data between runs.

15.1 Run `f1ad` — yesterday

Horizon	Best Model	RMSE	MAE	R ²
Day+1	Random Forest	19.32	15.62	-1.729
Day+2	Gradient Boosting	16.80	12.54	-0.559
Day+3	Gradient Boosting	19.69	14.80	-1.100

Table 16: Run `f1ad` — Day 2.

15.2 Run `f1ad` — today

Horizon	Best Model	RMSE	MAE	R ²
Day+1	Random Forest	19.44	15.88	-1.596
Day+2	Gradient Boosting	11.76	8.76	0.196
Day+3	Gradient Boosting	19.69	14.80	-1.100

Table 17: Run `f1ad` — Day 1.

15.3 What Changed?

Metric	Yesterday	Today	Change
Day+2 RMSE	16.80	11.76	-5.04 (↓ improvement)
Day+2 MAE	12.54	8.76	-3.78 (↓ improvement)
Day+2 R^2	-0.559	0.196	+0.755 (↑ improvement)
Day+1 RMSE	19.32	19.44	+0.12 (slight increase)
Day+1 MAE	15.62	15.88	+0.26 (slight increase)
Day+3	Identical		

Table 18: Changes between runs — Day+2 metrics improved significantly as the spike entered the training set. Day+1 metrics slightly worsened due to a more difficult test window.

15.4 Why This Proves the System Is Working

The same code, on different days, produces different metrics because the test window shifted.

- **today's test window:** Contained less of the Week 21 AQI spike → Day+2 R^2 = +0.196 (positive)
- **yesterday's test window:** Contained more of the Week 21 spike → Day+2 R^2 = -0.559 (negative)

15.5 Bottom Line

This is exactly how a healthy time-series forecasting system behaves. Metrics fluctuate with test window difficulty. The underlying model quality is improving with each daily retraining run.

15.6 Summary of Observations

Mistake	Lesson
Celebrating early perfect metrics	Single test window can be lucky
<code>TimeSeriesSplit</code> with <1000 rows	CV needs 5,000+ rows for time series
Median imputation for missing weather	Fake data → fake patterns
Interpreting negative R^2 as model failure	Hard test sets produce honest metrics
Expecting stable metrics across runs	Time-series metrics fluctuate naturally

Table 19: Summary of key lessons learned.

15.7 DagsHub Reference Guide

The screenshots below map commit tags to experiments:

Tag	What It Represents	Key Observation
95c6 / c3a3	Initial baseline	Excellent metrics on easy test window
baa5	TimeSeriesSplit experiment	Catastrophic RMSE increase
a4a5	Median imputation attempt	Severe performance degradation
1c50	Excluding NaN records	Improvement, positive R square for 2 days
a954	Multiple manual runs	Same tag, different metrics (test window effect)
f1ad	Final stable state	Documented performance

Table 20: Mapping of commit tags to experiments for DagsHub reference.

16 Limitations and Future Work

16.1 Known Limitations

1. **Weather API historical depth:** Open-Meteo’s weather forecast API retains only ≈ 31 days of history, versus 90 days for the air quality API. This leaves 66% of stored records with null weather features, limiting the effective training set to rows after May 7, 2026. This is a structural API constraint, not a data pipeline bug.
2. **Dataset size:** With ≈ 763 complete training rows (at the time of writing), longer-horizon forecasts (Day+2, Day+3) are more sensitive to the composition of the test window. Metrics are expected to stabilise as the automated hourly pipeline accumulates data.
3. **Single-split evaluation:** The 80/20 temporal split produces metrics that are sensitive to the characteristics of the test window. TimeSeriesSplit cross-validation would provide more stable estimates but requires a minimum of $\approx 5,000$ rows to produce meaningful fold sizes.
4. **No recurrent deep learning:** The deep learning model is a feedforward MLP that treats each timestep independently. An LSTM or Transformer would leverage the temporal structure more effectively.

16.2 Future Work

- Introduce TimeSeriesSplit cross-validation once $\approx 5,000$ complete rows are available.
- Add cyclical hour encoding (sin/cos of hour) to better represent the continuous diurnal cycle.
- Extend lag windows to 6h, 12h, and 24h for richer temporal context.
- Implement an LSTM or Temporal Fusion Transformer for the deep learning component.

- Integrate additional data sources (traffic density, industrial event calendar) to improve spike prediction.

17 Conclusion

This project delivers a complete, production-grade AQI forecasting system for Karachi that satisfies all specified requirements: automated hourly data ingestion, feature engineering, daily model retraining across five model families spanning statistical to deep learning approaches, SHAP-based feature importance explanations, hazardous AQI alerts, and a publicly deployed interactive dashboard. The pipeline is fully autonomous — once deployed, it requires no manual intervention and self-improves daily as new data accumulates. The identified limitation regarding Open-Meteo’s weather API depth is documented transparently, and a clear path to improved metrics as data grows is established.